

Proposition 10.1 (Multifold binomial sums of powers). *For non-negative integers n, m , and an arbitrary integer t ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m. \end{aligned}$$

For example,

Example 10.2 (Examples of multifold binomial sums). *For integer $0 \leq t \leq 2$*

$$\begin{aligned} t = 0 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \end{aligned}$$

$$t = 1 : \quad \Sigma^r n^m = \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right.$$